An explicit genus-3 plane quartic over F_q for an elliptic curve over F_{q³}

2026-10-12. Conductor T-2267. **Structural stage diagnostic on toy fields.** A curve is built and its Frobenius checked. No discrete logarithm is solved, no index calculus is run, and no speedup is claimed. Protocol frozen before the first run: PROTOCOL.md. Run `20261013T0320Z`: records, environment, run report. This report as PDF: REPORT.pdf. Follows `research/ghs_cubic_cover_census_20261012/` (PR #1759), which counted these covers by genus without building them.

Question

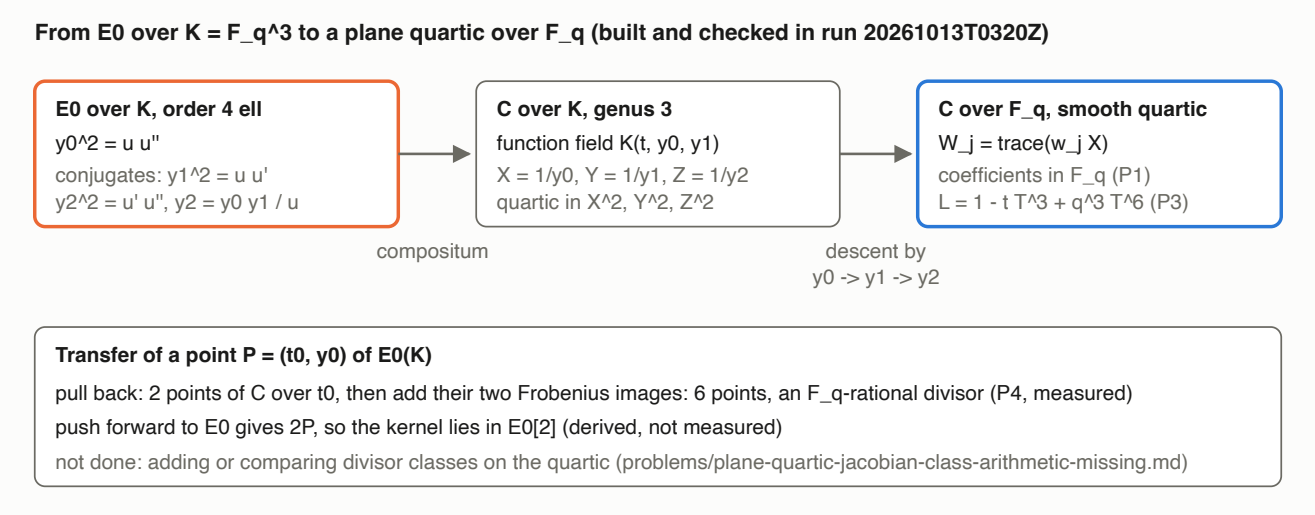
The census found that every isogeny class with `4 | #E(K)` over `K = F_{q³}` has a genus-3 cover by branch data. Does that cover exist as a curve over `F_q` whose Jacobian has `#E(K)` rational points, and is the transfer of points an explicit `F_q`-rational object?

Result

Yes on all nine cases built (`q ∈ {5, 7, 11}`, `#E0(K) = 4ℓ`). For each:

- the curve is a **smooth plane quartic with coefficients in `F_q`**, so it is not hyperelliptic;
- its point counts over `F_q`, `F_{q²}`, `F_{q³}` give L-polynomial `1 − t_E T³ + q³T⁶`, hence **`#Jac(C)(F_q) = #E0(K)`**;
- every transfer divisor tested is an `F_q`-rational divisor on it.

So the census of PR #1759 counted objects that exist as assumed. What is still not shown by execution is the transfer as a group homomorphism, because no class arithmetic on a plane quartic is implemented (problem record).



Construction (derived; two steps in Lean)

`u ∈ K[t]` monic quadratic, `u' = u^σ`, `u'' = u^{σ²}`, six distinct roots, `u, u', u''` linearly independent.

1. `E0 : y0² = u·u''`, `E1 = E0^σ : y1² = u·u'`, `E2 : y2² = u'·u''` with `y2 = y0·y1/u` (`DescentCycle.third_cover`).
2. `σ~ : y0 ↦ y1 ↦ y2 ↦ y0`, `t ↦ t`, `σ` on `K`, has order 3 (`DescentCycle.cycle_closes`), so the compositum descends to `F_q`.
3. With `X = 1/y0, Y = 1/y1, Z = 1/y2`: `(u : u' : u'') = (Z² : X² : Y²)`. If `(1, t, t²)ᵀ = R·(u, u', u'')ᵀ` then `t·t = 1·t²` reads `G = (R₁·w)² − (R₀·w)(R₂·w) = 0, w = (Z², X², Y²)`: a quartic in `X², Y², Z²`. The three differentials `dt/y_i` are holomorphic and `(X : Y : Z)` is the canonical map, so a smooth `G` means `C` is not hyperelliptic.
4. `σ~` permutes `X → Y → Z`. In `W_j = ω_j X + ω_j^σ Y + ω_j^{σ²} Z, ω = (1, g, g²)`, the quartic has `F_q` coefficients up to a scalar.
5. Transfer `φ = (1 + Fr + Fr²)∘π0^*`, with `Fr(t, y0, y1, y2) = (t^q, y2^q, y0^q, y1^q)`. `π_j_* ∘ π_0^* = 0` for `j ≠ 0` (it factors through the `t`-line), so `π0_* ∘ φ = [2]` and `ker φ ⊂ E0[2]`. With `#Jac(C)(F_q) = #E0(K) = 4ℓ`, `φ` maps `E0(K)[ℓ]` isomorphically onto `Jac(C)(F_q)[ℓ]`.

Report statement	Lean declaration	Status
`(u·y2)² = u²·u'·u''`	`third_cover`	proved, any commutative multiplication
`u·(y1·y2) = (u·u')·y0` (cycle closes)	`cycle_closes`	proved
cancelling `u`; the function field; descent; step 5	none	**not formalized**

Lean: lean/DescentCycle.lean, toolchain v4.34.1, no Mathlib, no axioms. No identity certificate: the report's identities are conditional on the curve equations, not polynomial identities.

Measurements (run 20261013T0320Z, exact counts)

Tool: genus3_build.rs. `u = t² + bt + c` in lexicographic order of field index; the first cases with `#E0(K) = 4ℓ`. `N_k` are projective point counts of the `F_q` quartic over `F_{q^k}` by exhaustion; `N3` was also counted on the `K` model and agreed in all nine cases.

q	u (index of b, c)	roots of u in K	#E0(K)	t_E	in F_q	smooth	N1, N2, N3	predicted	#Jac(C)(F_q)
5	5, 30	0	148 = 4·37	−22	yes	yes	6, 26, 192	6, 26, 192	148
5	5, 51	2	148 = 4·37	−22	yes	yes	6, 26, 192	6, 26, 192	148
7	7, 52	0	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 54	0	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 110	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 118	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 129	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
11	11, 130	0	1348 = 4·337	−16	yes	yes	12, 122, 1380	12, 122, 1380	1348
11	11, 172	2	1324 = 4·331	8	yes	yes	12, 122, 1308	12, 122, 1308	1324

Example, `q = 7`, u index (7, 110), over `F_7`:

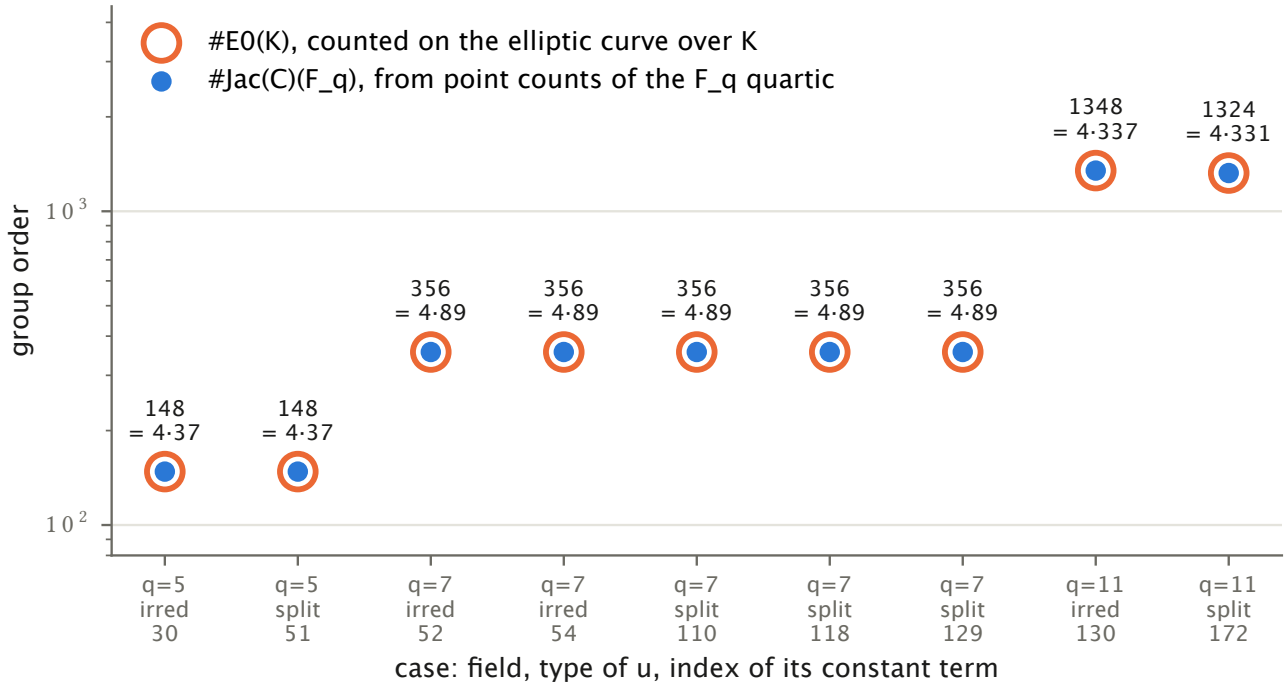
```
W2⁴ + W1²W2² + 6W1³W2 + 6W1⁴ + 4W0W2³ + W0W1W2² + 4W0W1²W2 + 4W0W1³
  + W0²W2² + 4W0²W1W2 + 5W0²W1² + 6W0³W2 + 6W0⁴ = 0
```

All nine quartics are in the records (`quartic_Fq`, entries `[i, j, k, coefficient]` for `W0^i W1^j W2^k`).

Transfer divisors (P4). For each `P = (t0, y0) ∈ E0(K)` with `u·u'(t0)` a non-zero square, the two points of `C` over `P` and their two Frobenius images each:

q	u	points tested	on the F_q quartic	q-power = model Fr	image on E0	points of E0 not tested
5	5, 30	564	564	564	564	52
5	5, 51	564	564	564	564	48
7	7, 52 / 7, 54	1128 each	1128	1128	1128	166 each
7	7, 110 / 118 / 129	1092 each	1092	1092	1092	168 each
11	11, 130	4128	4128	4128	4128	658
11	11, 172	3876	3876	3876	3876	672

The untested points are those where `u·u'(t0)` is not a square: their two preimages are conjugate over `F_{q⁶}` and need a field the tool does not build.



P1, P2, P3 and P4 hold on every case; no falsifier fired.

Scope and limits

- **Not executed:** φ as a homomorphism, its injectivity on `E0[ℓ]`, any discrete logarithm, any relation collection. These need class arithmetic on the quartic; see the problem record.
- **Selection.** The frozen rule takes the first cases in index order. All five `q = 7` cases landed in one isogeny class (`#E0(K) = 356`); the other 4ℓ class at `q = 7` seen in the census (332) was not built. Most candidates were skipped as linearly dependent (2394 of 2443 at `q = 7`) because `b ∈ F_q` comes first in index order.
- **One construction.** Only the `m = 2, r = 6` type. The hyperelliptic `m = 3, r = 5` type is not built.
- **Cost (published asymptotics, not measured):** index calculus on a smooth plane quartic over `F_q` is reported at about `q` operations (Diem), against about `q^{3/2}` for rho on `E(K)`. These toy fields are far too small to compare anything.
- `n = 3`, `q ∈ {5, 7, 11}`, odd characteristic above 3.

Next discriminating step

Implement linear equivalence on a plane quartic and execute the transfer on the `q = 7` case (7, 110); then relation collection by lines through pairs of `F_q`-points, counted against rho. The acceptance conditions are in the problem record.

Graphs checked and left unchanged

No canonical graph changes (no ECDLP rung, no timing; no `benchmarks/` entry).

Reproduce

```
rustc --edition 2021 -O genus3_build.rs && ./genus3_build 5 1 1 && ./genus3_build 7 3 2 && ./genus3_build
11 1 1
python3 reports/plot.py results/run-20261013T0320Z.jsonl reports/run-20261013T0320Z   # layout only
cd lean && lake build
```